

Datasheet

Artem Dunaievskiy, Sanya Ahmad, Jake Briscoe, Jamie Turbyne
(ad2955 sa3581 jb3802 jt666)

Product Information:

Name: Modular Display for Weather Conditions

Version: V1.0 (28/04/2025)

General Description:

Description:

The device is based on the principle of a Modular Display and an Arduino ESP32 Nano board [1]. By using six high-brightness LEDs and LED control circuits based on PCB, the device can display weather information on six different acrylic sheets. The Arduino board uses data from the OpenWeatherMap API [2] to retrieve the necessary information and control the circuits using the associated pins. This is a prototype version.

Applications:

- Smart weather monitoring
- Recommendation for home or demonstrations applications

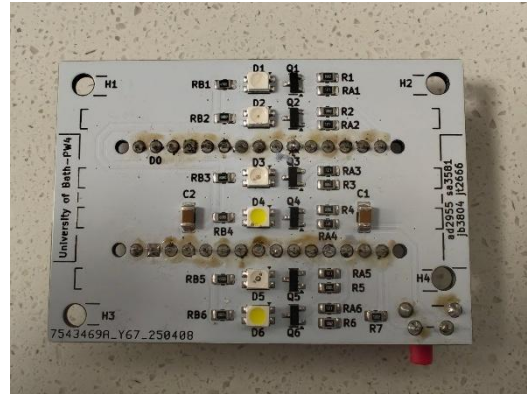


Figure 1 Assembled PCB

Features:

- Arduino Nano ESP32-based design
- PCB based
- 6-segment modular display
- Ergonomic structure for ease of use
- Fully modular system for customizable configurations
- High-brightness LEDs for clear visibility
- Wireless connectivity based on Wi-Fi

Table of Contents:

Product Information:	1
General Description:.....	1
Table of Contents:	2
Specifications and characteristics	3
Electrical specification:	3
Mechanical specification:.....	3
Environmental specification:	4
Embedded electronics.....	5
Key Performances of the Device:	5
Interfaces and connections:	5
Diagrams of the device:	6
Bill of Materials:	7
Pin Configuration:	8
Change Log:.....	8
References:.....	9

Specifications and characteristics

Electrical specification:

(based on Arduino datasheet [1])

Table 1 Electrical Characteristics

Description	Value	Unit
Arduino I/O Voltage	3.3	V
Arduino Input voltage (nominal)	6-21	V
Arduino Source Current per I/O Pin	40	mA
Arduino Sink Current per I/O Pin	28	mA
Device Power Supply Track	5	V

Table 2 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
V USB	Input voltage from USB connector	4.8	5.0	5.5	V
T ambient	Ambient Temperature	-40	25	105	°C

Mechanical specification:

Table 3 Dimensions of Device

Name	Value	Unit
Length	60.96	mm
Width	41.91	mm
Hight (without acrylic sheets and holder)	20	mm

Table 4 Dimensions of Acrylic Sheet

Name	Value	Unit
Width	53.3	mm
Height	39.3	mm
Thickness	3	mm

Mounting Options:

- 4 M3 mounting holes on the PCB
- Dedicated mounting zone on the PCB

Elements of system Type/Material:

- MDF Acrylic Layers Holder
- Acrylic Sheets
- PCB



Figure 2 Acrylic sheets and Holder

Environmental specification:

Arduino Environmental Standards:

- RoHS [3] - ensuring no hazardous materials like lead, mercury, cadmium, hexavalent chromium, PBB, and PBDE are present above safe thresholds
- REACH [4] - ensuring no Substances of Very High Concern (SVHC) beyond allowable limits
- FCC [5] – ensuring that the device does not cause harmful interference and can withstand reasonable levels of external interference
- UKCA [6]- ensuring health, safety, and electromagnetic compatibility standards are met for the UK market

Device Environmental Features:

- Modular design – allowing easy disassembly of the device into various parts, resulting in simpler reuse and recycling of components
- Upgrade possibilities – the device features the ability to replace broken parts, due to its modular structure and detailed documentation, thereby reducing overall e-waste

Embedded electronics

Key Performances of the Device:

Table 5 Key Performances of the Device

Feature	Description	Value	Unit
Microcontroller	NORA-W106-10B (ESP32-S3 SoC)		
Microprocessor	Xtensa Dual-core 32-bit LX7:		
	Frequency	Up to 240	MHz
	ROM	384	kB
	SRAM	512	kB
Memory	PSRAM	8	MB
Connectivity	Wi-Fi® 4 IEEE 802.11(b/g/n):	802.11b: 11 802.11g: 54 802.11n: 72	Mbit/s
	Bluetooth® LE v5.0 data rates:	up to 2	Mbps
LEDs	White LED Luminous Intensity	1400	mcd
	Green LED Luminous Intensity	5000	mcd
	Red LED Luminous Intensity	3000	mcd
	Yellow LED Luminous Intensity	2000	mcd
	Blue LED Luminous Intensity	1000	mcd

(all relevant information included in Bill of Materials Table 6)

Interfaces and connections:

Input Interfaces and Specification:

- USB connector - USB-C
- Wi-Fi - u-blox® NORA-W106 (ESP32-S3)
- Tactile Switch - [7]

Output Interfaces:

- Modular display

Diagrams of the device:

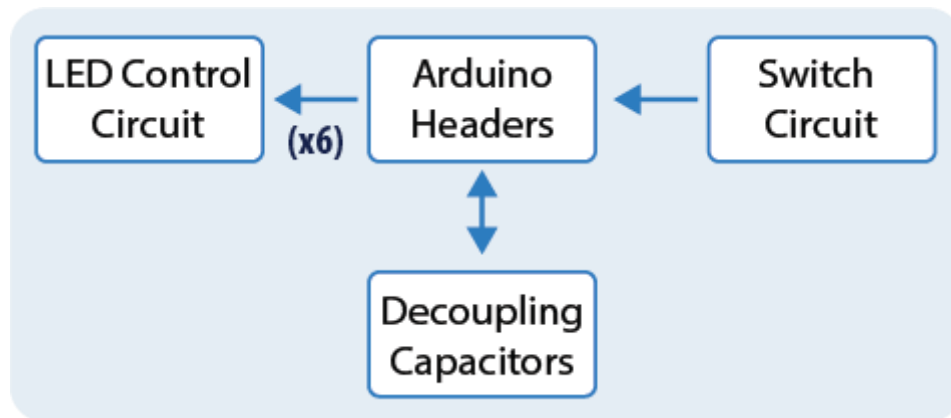


Figure 3 Block Diagram

(Reference:[8])

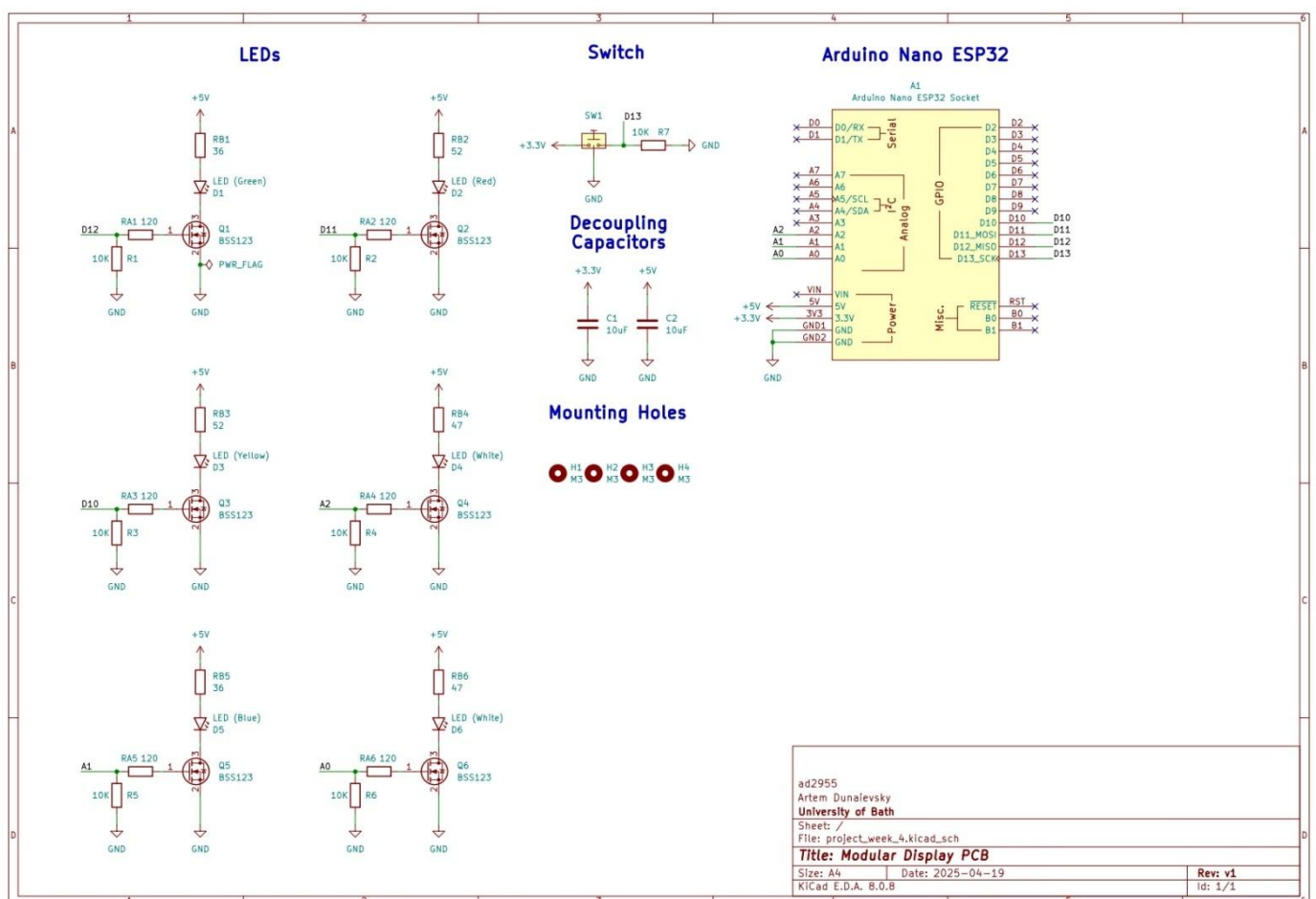


Figure 4 PCB Schematic

Bill of Materials:

Table 6 Bill of Materials:

N	References	Value	Datasheet	Description	Farnell Part Number	Manufacturer	Price/Unit [£]	Quantity
1	C1, C2	10uF	https://www.farnell.com/datasheets/3626220.pdf	Unpolarized capacitor	3881062	MULTICOMP PRO	2.07	2
2	R1, R2, R3, R4, R5, R6, R7	10K	https://www.farnell.com/datasheets/2563624.pdf	Resistor	2447553	MULTICOMP PRO	0.08	7
3	D4, D6	LED (White)	https://www.farnell.com/datasheets/77929.pdf	LED (White)	1649009	CREE LED	0.87	2
4	D1	LED (Green)	https://www.wonline.com/components/products/datasheet/150141GS73140.pdf	LED (Green)	3619644	WURTH ELEKTRONIK	0.32	1
5	D2	LED (Red)	https://www.wonline.com/components/products/datasheet/150141RS73140.pdf	LED (Red)	3619649	WURTH ELEKTRONIK	0.28	1
6	D3	LED (Yellow)	https://www.wonline.com/components/products/datasheet/150141YS73140.pdf	LED (Yellow)	3619650	WURTH ELEKTRONIK	0.32	1
7	D5	LED (Blue)	https://www.wonline.com/components/products/datasheet/150141BS73140.pdf	LED (Blue)	3619646	WURTH ELEKTRONIK	0.32	1
8	SW1	~	https://www.farnell.com/datasheets/3457679.pdf	Tactile switch	3796560	APEM	1.55	1
9	A1	Arduino Socket	https://www.farnell.com/datasheets/2925583.pdf?_gl=1*1eh5gke*_gcl_au*MjY4NzkwMTMzLjE3NDQ5ODYxNzC_	Arduino Nano ESP32 Socket	2212S-24SG-85	MULTICOMP PRO	1.52	1
10	Q1, Q2, Q3, Q4, Q5, Q6	BSS123	http://www.diodes.com/assets/Datasheets/ds30366.pdf	0.17A Id, 100V Vds, N-Channel MOSFET, SOT-23	1510764	NEXPERIA	0.99	6
11	RA1, RA2, RA3, RA4, RA5, RA6	120	https://www.farnell.com/datasheets/2860635.pdf	Resistor	2073628	MULTICOMP PRO	0.12	6
12	RB1, RB5	36	https://www.farnell.com/datasheets/2563624.pdf?_gl=1*1owqza8*_gcl_au*MjY4NzkwMTMzLjE3NDQ5ODYxNzC_	Resistor	2447508	MULTICOMP PRO	0.14	2
13	RB2, RB3	52	https://www.farnell.com/datasheets/2189593.pdf?_gl=1*1o8lqk4*_gcl_au*MjY4NzkwMTMzLjE3NDQ5ODYxNzC_	Resistor	2694892	MULTICOMP PRO	0.04	2
14	RB4, RB6	47	https://4donline.ihs.com/images/VipMasterIC/IC/WATC/WATCS00985/WATCS00985-1.pdf?hkey=6D3A4C79FDBF58556ACFDE234799DDF0	Resistor	2670099	MULTICOMP PRO	0.16	2

Pin Configuration:

Table 7 Pin Configuration

Pin number	Led color	Figure and Description
D12	Green	Happy Face - Good Weather
D11	Red	Sad Face - Bad Weather
D10	Yellow	Sun - Sunny Weather
A2	White	Cloud - Cloudy Weather
A1	Blue	Snowflake - Snowy Weather
A0	White	Wind - Windy Weather
D13	N/A	Switch

Change Log:

Table 8 Change Log

Date	Changes
28/04/2025	Prototype

References:

- [1] 'Nano ESP32 | Arduino Documentation'. Accessed: Apr. 27, 2025. [Online]. Available: <https://docs.arduino.cc/hardware/nano-esp32/>
- [2] 'Current weather and forecast - OpenWeatherMap'. Accessed: Apr. 27, 2025. [Online]. Available: <https://openweathermap.org/city/2643743>
- [3] 'RoHS Directive - European Commission'. Accessed: Apr. 28, 2025. [Online]. Available: https://environment.ec.europa.eu/topics/waste-and-recycling/rohs-directive_en
- [4] 'REACH Regulation - European Commission'. Accessed: Apr. 28, 2025. [Online]. Available: https://environment.ec.europa.eu/topics/chemicals/reach-regulation_en
- [5] 'Equipment Authorization | Federal Communications Commission'. Accessed: Apr. 28, 2025. [Online]. Available: <https://www.fcc.gov/engineering-technology/laboratory-division/general/equipment-authorization>
- [6] 'Using the UKCA marking', GOV.UK. Accessed: Apr. 28, 2025. [Online]. Available: <https://www.gov.uk/guidance/using-the-ukca-marking>
- [7] 'Tact switches • 6 x 6 mm standard package'. [Online]. Available: <https://www.farnell.com/datasheets/3457679.pdf>
- [8] L. Jabban and J. Graham-Harper-Cater, 'Modular Display Project - PCB Schematic - EE12001'. Accessed: Apr. 26, 2025. [Online]. Available: https://moodle.bath.ac.uk/pluginfile.php/2606566/mod_resource/content/31/_site/PCB-Assembly-and-Design/301-modular-display-project-pcb-schematic.html